

Notes: Siddique, Vlassopoulos & Zenou (QJE, 2026)
Leveraging Virtual Contact and Social Networks to Foster Interethnic Harmony

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1 Big picture

- **Question.** Can virtual contact through a documentary film improve interethnic harmony in villages with longstanding ethnic tension?
- **Setting.** Multiethnic villages in northwestern Bangladesh, where ethnic-majority Bengalis and ethnic-minority Santals have histories of social distance, land conflict, and discrimination.
- **Intervention.** A documentary portrays Santal lives, culture, hardships, and success stories. Control participants watch a neutral documentary on flower farming.
- **Design.** 121 villages are cluster-randomized into control, random-targeting, and network-central-targeting arms.
- **Main result.** Treated Bengalis show stronger prosocial behavior toward Santals, especially altruism and trust. Measured stereotypes and discriminatory opinions do not change robustly.
- **Network result.** In villages where network-central individuals are targeted, untreated Bengalis and Santals also show behavioral improvements.
- **Economic takeaway.** Virtual contact can produce behavioral improvements in interethnic relations, and network targeting may amplify those effects.

2 Data and measurement

2.1 Village sample and treatment arms

The study uses 121 multiethnic villages in the Rajshahi division of Bangladesh. Villages are assigned to control, random treatment, or central treatment. In random villages, randomly selected Bengalis watch the Santal documentary. In central villages, high-diffusion-centrality Bengalis and randomly selected Bengalis watch it. Control viewers watch a flower-farming documentary.

Interpretation. The design contrasts Santal-focused virtual contact with a neutral media experience and separately tests whether treating central individuals changes village-level effects.

2.2 Outcomes

The main outcomes are behavioral games measuring altruism, solidarity, and trust; self-reported interactions and visits; stereotypes and discriminatory opinions; emotion measures during film viewing; administrative complaints; and productivity in a later casual-work field experiment.

Interpretation. The strongest evidence comes from incentivized behavior, administrative data, and real-effort production rather than only self-reported attitudes.

3 Models and empirical specifications

3.1 Main treatment-effect regression

$$Y_{ijc} = \alpha + \beta_1 RR_{ijc} + \beta_2 RC_{ijc} + \beta_3 CC_{ijc} + \zeta Y_{0ijc} + \Gamma X_{ijc} + \nu_c + \varepsilon_{ijc}.$$

RR and RC are randomized treatment comparisons. CC is descriptive because central individuals are selected by network position.

Interpretation. The regression estimates direct treatment effects on treated Bengalis while controlling for baseline outcomes, selected covariates, and union-council fixed effects.

3.2 Untreated spillover specification

$$Y_{1ijc} = \alpha + \delta_1 \text{RandomVillage}_j + \delta_2 \text{CentralVillage}_j + \zeta Y_{0ijc} + \Gamma X_{ijc} + \nu_c + \varepsilon_{ijc}.$$

Interpretation. This specification asks whether untreated individuals in treated villages respond, and whether central targeting produces stronger village-level effects than random targeting.

3.3 Inference and multiple hypotheses

The paper clusters standard errors at the village level, reports randomization-inference p-values, and uses Westfall–Young family-wise-error adjustments within preregistered outcome families.

Interpretation. Multiple-testing corrections are essential because the paper measures many behavioral, belief, administrative, and productivity outcomes.

4 Identification and empirical design

4.1 Cluster randomization and causal comparisons

Village-level randomization limits within-village contamination and matches the social-spillover level. RR and RC are clean causal comparisons. CC is descriptive because centrality is selected.

Interpretation. The cleanest causal claims are for randomly selected treated Bengalis and randomized village-level spillovers.

4.2 Central targeting

The central arm targets people nominated as effective spreaders of information. Stronger effects among untreated people in central villages indicate that who is treated matters.

Interpretation. The design tests a network-policy question, but it cannot isolate whether diffusion occurs through conversation, visibility, prestige, or norm formation.

4.3 Measurement and robustness

The paper combines incentivized games, surveys, emotion detection, administrative complaints, and a later work experiment. It controls for baseline outcomes and LASSO-selected covariates and reports RI and FWER p-values.

Interpretation. The evidence is strongest when multiple outcome types point in the same direction.

5 Tables and figures

5.1 Figure I: Prosocial Outcomes at End Line; Table I: Treatment Effects on Behaviors of the Treated Ethnic Majority

Read: Figure I shows raw means for altruism, solidarity, and trust among treated Bengalis. Table I adds regression-adjusted estimates and multiple-testing adjustments.

Economic message. The documentary shifts observed prosocial behavior more strongly than broad self-reported contact.

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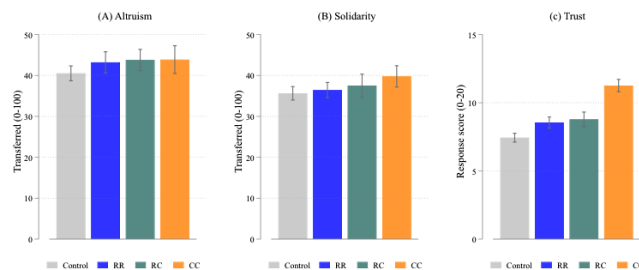


FIGURE I
Prosocial Outcomes at End Line

The sample includes treated Bengalis in the two treatment arms and the control arm. Bars indicate 95% confidence intervals. Abbreviations: RR (Random-in-Random), RC (Random-in-Central), and CC (Central-in-Central). All comparisons shown are unconditional (i.e., with no covariate adjustment), and standard errors are clustered at the village level. All p -values below are unadjusted two-sided tests on raw means. Panel A: control versus RR: $p = .10$; control versus RC: $p = .05$; control versus CC: $p = .09$; RR versus RC: $p = .76$; RR versus CC: $p = .76$; RC versus CC: $p = .96$. Panel B: control versus RR: $p = .52$; control versus RC: $p = .26$; control versus CC: $p = .01$; RR versus RC: $p = .54$; RR versus CC: $p = .04$; RC versus CC: $p = .22$. Panel C: control versus RR: $p < .0001$; control versus RC: $p < .0001$; control versus CC: $p < .0001$; RR versus RC: $p = .49$; RR versus CC: $p < .0001$; RC versus CC: $p < .0001$.

Figure I: Prosocial outcomes at end line

TABLE I
TREATMENT EFFECTS ON BEHAVIORS OF THE TREATED ETHNIC MAJORITY

Variables	Prosociality			Self-reports				Interactions and visits			
	Altruism Solidarity (1)	Trust (2)	Ex sol (3)	Friends (4)	Water\$ (5)	Help (7)	Int San (8)	Int Ben (9)	Visit San (10)	Visit Ben (11)	
RR	2.593 (2.218)	2.525* (1.475)	0.878*** (0.318)	0.451 (1.094)	-0.096 (0.117)	-0.849* (0.417)	0.420*** (0.074)	0.023 (0.015)	0.007 (0.034)	1.231 (1.376)	
RC	5.966*** (2.076)	2.867 (1.869)	1.617*** (0.454)	3.027 (2.403)	0.138 (0.162)	-1.633 (1.031)	0.253 (0.186)	0.012 (0.082)	-0.025 (0.022)	0.869 (1.110)	
CC	6.542*** (2.336)	6.178*** (1.755)	4.063*** (0.433)	-0.527 (2.253)	0.059 (0.152)	-0.566 (0.907)	0.432** (0.204)	0.044 (0.086)	0.011 (0.020)	1.192 (1.026)	
Observations	1,516	1,515	1,513	1,511	1,519	444	1,394	1,515	1,515	1,514	
Control mean	40.52	35.63	7.437	29.84	1.881	0.521	6.575	1.789	3.985	19.30	
Control std. dev.	23.77	21.95	3.796	22.62	1.712	5.103	2.945	0.932	0.163	12.80	
RR = RC p -values	.13	.85	.09	.29	.14	.28	.30	.89	.14	.72	
RR = CC p -values	.12	.11	<.001	.67	.29	.69	.95	.81	.87	.97	
RC = CC p -values	.77	.19	<.001	.08	.58	.26	.24	.73	.15	.77	
RR + RC vs. C p -values	.08	.06	<.01	.46	.80	.09	<.01	.77	.90	.16	
FWER p -values on RR	.11	.04	<.01	.97	.90	.97	.05	.99	.95	.65	
FWER p -values on RC	<.01	.11	<.001	.64	.84	.88	.64	.99	.51	.89	

TABLE I
CONTINUED

Variables	Prosociality			Self-reports				Interactions and visits			
	Altruism (1)	Solidarity (2)	Trust (3)	Ex sol (4)	Friends (5)	Water\$ (6)	Help (7)	Int San (8)	Int Ben (9)	Visit San (10)	Visit Ben (11)
FWER p -values on CC	<.001	<.01	<.001	.97	.97	.97	.37	.91	.99	.60	.99
RR p -values on RR	.26	.67	<.01	.69	.45	.82	<.01	.72	.50	.21	.92
RR p -values on RC	.01	.20	<.01	.20	.30	.44	.20	.74	.16	.35	.65
RR p -values on CC	<.01	<.01	<.001	.98	.60	.83	.08	.37	.87	.18	.68

Notes: The sample includes treated Bengalis in both treatment arms and the control arm. RR is the treatment arm where participants were selected randomly and are from the random arm. RC is the sample where participants were selected randomly and are from the control arm. CC is the sample where all participants had high distance centrality, were selected using the group method, and are from the control arm. All specifications include univariate fixed effects, the outcome measured at baseline, education, income, and other controls selected using the post-double robust LASSO (Hothorn, Hornik, Strobl, and Zeileis 2015). Outcomes in the columns are as follows: column (1) Altruism or the amount (0-100) transferred to a minority measured using the dictator game (Fehr and Gächter 1998); column (2) Solidarity or the amount (0-100) transferred to a minority, conditional on a risk shock, measured using a simplified solidarity game (Fehr and Gächter 1998); column (3) Trust is the level of trust toward minorities measured using a slightly modified Global Preference Survey question (Falk et al. 2014), which was answered on a 0-20 thermometer scale; column (4) Ex sol is the solidarity expected from the respondent (0-100); column (5) Friends is the number of minority friends among the 10 closest friends; column (6) Water\$ is the amount charged to minorities in RHP when they come to fetch water from one well; column (7) Help is the participant's willingness to help others (scored on a scale between 0 and 10, where 10 means very willing to help); column (8) and (9) social interactions with non-coethnic and coethnic, column (10) and (11) frequency of visits to non-coethnic and coethnic, house. FWER p -values correspond to multiple hypothesis testing-adjusted p -values computed using the Westfall and Young (1993) correction. To compute the FWER-adjusted p -values, we consider prosociality, self-reported behaviors, and interactions and visits as three families of outcomes. RE p -values correspond to randomization inference-based p -values computed following Young (2013). We used OLS regressions (with all controls available), instead of plim to compute the FWER and RE p -values. We also report the control group's means and standard deviations. RR = RC versus C p -values report the p -values of the point effects of RR and RC versus C in the control group. Robust standard errors in parentheses are clustered at village level. * $p < .1$, ** $p < .05$, *** $p < .01$.

5.2 Figure II: Treatment Effects Across Domains; Table II: Impacts on Beliefs of Treated Ethnic Majority

Read: Figure II summarizes domain-level effects. Table II shows limited evidence of changes in stereotypes or discriminatory opinions.

Economic message. The mechanism appears more affective and behavioral than cognitive.

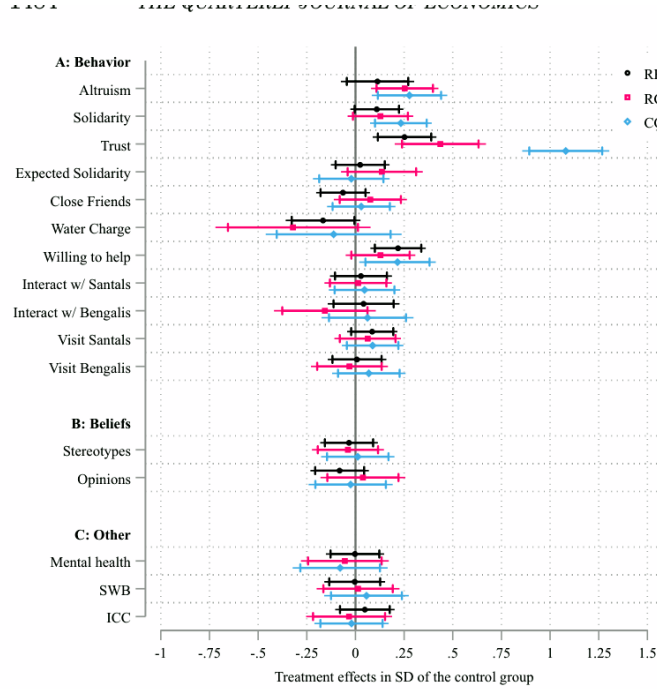


FIGURE II
Treatment Effects on Behavior, Beliefs, and Other Outcomes of the Treated Ethnic-Majority Sample

This figure reports treatment effects in standard deviation units of the control group, with 90% and 95% confidence intervals. All outcomes have been control-group standardized

Figure II: treatment effects across domains

TABLE II IMPACTS ON BELIEFS OF TREATED ETHNIC MAJORITY							
Variables	Panel A: Stereotypes						
	Stereotype index (1)	Useless (2)	Doctors (3)	Establish (4)	Teachers (5)	Education (6)	Agriculture (7)
RR	-0.083 (0.070)	0.216 (0.148)	0.258*** (0.062)	0.150 (0.111)	-0.077 (0.251)	-0.100 (0.176)	-0.628** (0.297)
RC	-0.039 (0.094)	0.005 (0.187)	-0.028 (0.089)	-0.004 (0.119)	0.099 (0.368)	-0.284 (0.227)	-0.127 (0.371)
CC	0.012 (0.096)	0.150 (0.152)	0.012 (0.094)	-0.017 (0.120)	0.121 (0.163)	-0.024 (0.236)	-0.035 (0.372)
Observations	1,476	1,509	1,509	1,496	1,502	1,498	1,485
Control mean	0	2.738	3.241	3.875	4.800	3.330	5.530
Control std. dev.	1	2.024	1.616	1.376	3.459	3.584	3.993
RR = RC p-value	.85	.84	< .01	.88	.61	.37	.58
RR = CC p-value	.62	.71	< .01	.39	.55	.74	.10
RC = CC p-value	.56	.41	.41	.50	.92	.36	.79
RR = RC versus CC p-value	.62	.27	< .01	.29	.90	.43	.07
FWER p-value on RR	1.00	.56	< .001	.58	1.00	1.00	.30
FWER p-value on RC	1.00	1.00	1.00	1.00	1.00	.93	1.00
FWER p-value on CC	1.00	1.00	1.00	1.00	1.00	1.00	1.00
RI p-value on RR	.71	.13	< .001	.14	.79	.64	.04
RI p-value on RC	.58	.89	.73	.57	.84	.32	.74
RI p-value on CC	.39	.48	.97	.87	.71	.85	.35

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Variables	Panel B: Discriminatory opinions						
	Discriminatory index (1)	Harvest (2)	Bar (3)	Schools (4)	Teaching (5)	Working (6)	Child's life (7)
RR	-0.081 (0.076)	0.029 (0.063)	-0.057 (0.153)	-0.064 (0.237)	0.001 (0.108)	-0.185 (0.234)	0.143* (0.079)
RC	0.038 (0.111)	0.013 (0.077)	-0.004 (0.170)	-0.035 (0.327)	0.048 (0.154)	0.218 (0.323)	-0.160 (0.135)
CC	-0.025 (0.110)	0.171** (0.078)	-0.175 (0.185)	-0.302 (0.320)	-0.092 (0.165)	-0.206 (0.323)	-0.161 (0.121)
Observations	1,485	1,513	1,511	1,511	1,505	1,498	1,501
Control mean	0	4.282	7.368	5.019	5.096	5.247	5.964
Control std. dev.	1	0.885	1.927	3.162	1.618	2.908	1.427
RR = RC p-value	.35	.84	.73	.82	.72	.31	.86
RR = CC p-value	.58	.07	.49	.84	.57	.41	.09
RC = CC p-value	.51	.02	.35	.91	.42	.66	.30
RR = RC versus CC p-value	.51	.73	.77	.80	.80	.79	.45
FWER p-value on RR	.87	1.00	1.00	1.00	1.00	.72	.39
FWER p-value on RC	1.00	1.00	1.00	1.00	1.00	1.00	.91
FWER p-value on CC	1.00	1.00	.30	.97	1.00	1.00	.79
RI p-value on RR	.22	.87	.74	.77	.98	.47	.07
RI p-value on RC	.98	.91	.98	.84	.91	.87	.32
RI p-value on CC	.88	.88	.38	.89	.55	.70	.71

Notes: The sample includes treated Bengalis in both treatment arms and the control arm. See the note for Table I for specification details and p-value definitions. All other information is as previously described. In this table, only the index outcomes (columns 1) have been control-group standardized, where the mean of the control group is zero and the standard deviation is one. Outcomes in the other columns are as follows: Panel A, column (2) Santals are often useless; column (3) Santals would not make good doctors; column (4) I do not know any Santals who have established themselves; column (5) Santals do not make very good school teachers; column (6) Santals do not continue beyond school; column (7) Santals should continue working in the agricultural sector. Panel B, column (1) All Santals I know are not honest people; column (2) I never met and do not know anyone by Santals; column (3) schools should be separate for Santals and Bengalis; column (4) there should not be more Santal teachers in my child's school; column (5) I do not enjoy working/doing business with Santals; column (7) none of my child's best friends are Bengalis; column (8) one can only trust a Santal (for presentation purposes, we report this here not re-estimated). Robust standard errors clustered at the village level are in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

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5.3 Figure III: Facial Expressions and Emotional Responses During Film Viewing

Read: Figure III shows that the Santal documentary triggers stronger emotional engagement, especially sadness, relative to the control documentary.

Economic message. The figure supports the empathy/affect mechanism.

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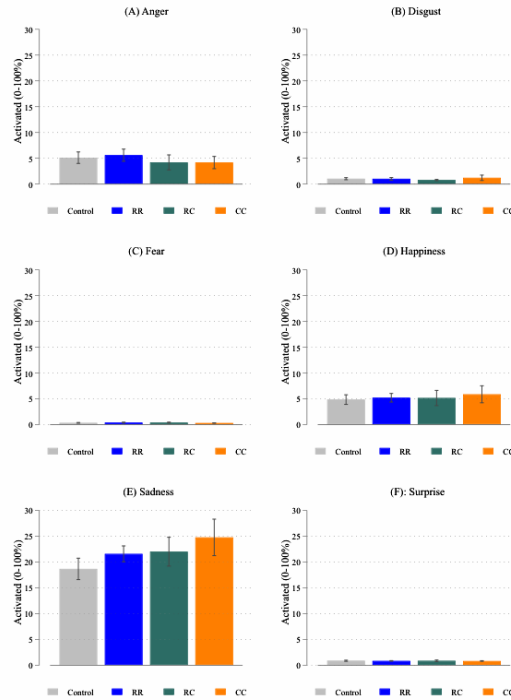


FIGURE III

Raw Differences in Emotions of Treated Ethnic Majority

Figure III: emotion measures during screening

5.4 Table III: Treatment Effects on Outcomes of the Untreated Ethnic Majority

Read: Table III shows stronger behavioral responses among untreated Bengalis in central villages.

Economic message. Targeting network-central people may amplify social effects among people who never watched the film.

5.5 Table IV: Treatment Effects on Behaviors of the Untreated Ethnic Minority

Read: Table IV shows responses among untreated Santals, including changes in trust and visits.

Economic message. Improved majority behavior can feed back into minority behavior and social expectations.

TABLE III
TREATMENT EFFECTS ON OUTCOMES OF THE UNTREATED ETHNIC MAJORITY

Variables	Panel A: Behaviors									
	Prosociality			Self-reports			Interactions and visits			
	Altruism (1)	Solidarity (2)	Trust (3)	Ex sol (4)	Friends (5)	Water8 (6)	Int San (7)	Int Ben (8)	Visit San (9)	Visit Ben (10)
Random (R)	1.245 (3.551)	-2.118 (4.099)	0.289 (0.297)	-3.370 (3.446)	0.212 (0.221)	-0.474 (0.323)	0.043 (0.112)	-0.013 (0.055)	1.187* (0.472)	1.386 (1.372)
Control (C)	6.979* (3.642)	9.656** (4.168)	0.479** (0.342)	-8.239 (6.198)	1.165*** (0.298)	-1.045* (0.454)	-5.140 (0.130)	0.124** (0.061)	0.094 (1.029)	2.790* (1.547)
Observations	800	800	797	794	800	214	800	799	799	800
Control mean	33.45	36.10	8.347	40.77	2.625	0.909	2.094	3.585	5.094	10.93
Control std. dev.	21.80	22.49	3.080	21.43	1.252	3.057	0.288	0.674	2.014	3.441
R = C <i>p</i> -values	.10	<.01	.27	.42	<.001	.23	.18	.01	.79	.35
FWER <i>p</i> -values on R	.89	.89	.89	.36	.39	.39	.81	.91	.06	.46
FWER <i>p</i> -values on C	.15	.01	.15	.39	.04	.39	.46	.03	.71	.11

Panel B: Beliefs		
Variables	Stereotypes (1)	Optimism (2)
Random (R)	0.004 (0.099)	-0.280* (0.146)
Control (C)	-0.031 (0.123)	0.526*** (0.192)
Observations	796	784
R = C <i>p</i> -values	.78	<.01
FWER <i>p</i> -values on R	.85	.15
FWER <i>p</i> -values on C	.95	<.01

Notes: Robust standard errors clustered at village level are in parentheses. The sample includes untreated Santals in the two treatment arms and in control villages. Random (R) denotes untreated Bengalis in random villages and Control (C) denotes untreated Bengalis in control villages. See the note for Table I for specification details and outcome definitions. For FWER adjustment, we treat prosociality, self-reported behaviors, and interactions and visits as three outcome families (Panel A), and stereotypes and discriminatory attitudes as two outcome families (Panel B). Panel B outcomes are standardized relative to the control group means and std. dev. one; for both indices, negative coefficients indicate more favorable views of non-coethnics. * $p < .1$, ** $p < .05$, *** $p < .01$.

TABLE IV
TREATMENT EFFECTS ON BEHAVIORS OF THE UNTREATED ETHNIC MINORITY

Variables	Prosociality			Self-reports			Interactions and visits			
	Altruism (1)	Solidarity (2)	Trust (3)	Ex sol (4)	Friends (5)	Water8 (6)	Int San (7)	Int Ben (8)	Visit San (9)	Visit Ben (10)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Random (R)	0.895 (4.508)	4.848 (4.198)	0.765** (0.343)	-3.695 (3.047)	0.358 (0.392)	0.507 (0.038)	-1.038** (0.522)	0.094 (0.396)	23.712*** (4.272)	18.717*** (4.454)
Control (C)	3.618 (4.583)	6.112 (5.000)	1.504*** (0.359)	4.365 (3.747)	-0.386 (0.428)	-1.859 (6.732)	-0.288 (0.577)	0.588 (0.422)	22.900*** (5.192)	10.584** (5.309)
Observations	721	721	717	720	721	93	720	719	720	721
Control mean	38.63	35.52	6.830	40.72	3.532	18.16	2.766	1.463	24.12	18.63
Control std. dev.	21.99	23.01	3.818	22.46	1.732	10.70	1.817	0.995	18.77	19.81
R = C <i>p</i> -values	.52	.72	<.01	.01	.03	.82	.12	.25	.82	<.01
FWER <i>p</i> -values on R	.57	.21	<.01	.55	.77	.85	<.01	.81	<.001	<.001
FWER <i>p</i> -values on C	.21	.12	<.001	.44	.82	.85	.48	.02	<.001	<.01

Notes: The sample includes untreated Santals in both treatment arms and the control arm. Random (R) denotes untreated Santals in random villages and Control (C) denotes untreated Santals in control villages. See the note for Table I for specification details, and *p*-value and outcome definitions. To compute the FWER-adjusted *p*-values, we consider prosociality, self-reported behaviors, and interactions and visits as three families of outcomes. All other information is as previously described. Robust standard errors in parentheses are clustered at the village level. * $p < .1$, ** $p < .05$, *** $p < .01$.

TABLE V
TREATMENT EFFECTS ON INTERETHNIC COMPLAINTS USING ADMINISTRATIVE DATA

Variables	To village counselors		To police stations	
	By Santal (1)	By Bengali (2)	By Santal (3)	By Bengali (4)
Random (R)	0.117 (0.278)	0.156 (0.480)	0.026 (0.230)	-0.130 (0.277)
Control (C)	0.102 (0.334)	0.063 (0.562)	0.082 (0.337)	-0.655** (0.326)
Control mean	1.05 [0.86]	1.72 [1.54]	0.90 [0.84]	1.58 [0.93]
Controls				
Village size	Y	Y	Y	Y
Village ethnic diversity	Y	Y	Y	Y
Complaints at baseline	Y	Y	Y	Y
Union Council fixed effects	Y	Y	Y	Y
R = C <i>p</i> -values	.96	.85	.86	.16

5.6 Table V: Treatment Effects on Interethnic Complaints Using Administrative Data

Read: Table V reports complaint counts to village counselors and police stations; the clearest effect is fewer Bengali police complaints in central villages.

Economic message. The intervention affects real village-level conflict behavior, not just survey responses.

5.7 Figure IV: Casual-Work Field Experiment

Read: Figure IV shows higher mixed-team output in central villages, driven by finishers.

Economic message. The intervention translates into productive cooperation in a real-effort labor task.

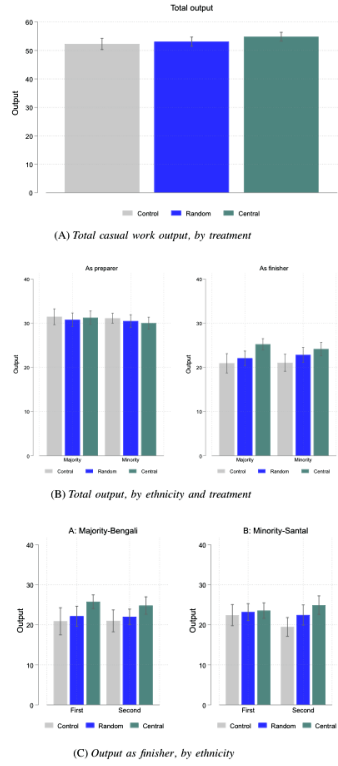


FIGURE IV

Casual-Work Field Experiment

The bars show mean output. Error bars report 95% confidence intervals with standard errors clustered at the village level. Comparisons are unconditional (no covariate adjustment). Panel A: Total output is preparer + finisher. Control versus random: $p = .53$; control versus central: $p = .05$; random versus central: $p = .13$. Panel B: Majority = Bengali, minority = Santal. Majority preparer: $p = .57, .86, .67$; minority preparer: $p = .50, .22, .63$; majority finisher: $p = .42, < .01, < .01$; minority finisher: $p = .17, .01, .25$ (control versus random, control versus central, random versus central). Panel C: “First” denotes assigned finisher at start; “second” denotes completing as finisher. Majority (first): $p = .56, .01, .02$; majority (second): $p = .55, .04, .06$; minority (first): $p = .65, .50, .81$; minority (second): $p = .09, < .01, .16$ (control versus random, control versus central, random versus central).

Figure IV: casual-work field experiment

6 Short summary

- The paper evaluates a Santal documentary as virtual intergroup contact.
- Treated Bengalis show improved prosociality but not robust belief changes.
- Emotional measures suggest an affective empathy channel.
- Central targeting creates stronger spillovers to untreated Bengalis and Santals.
- Administrative complaints and work-task productivity provide behavior beyond surveys.

7 Conclusion

7.1 Core conclusion

The paper shows that documentary-based virtual contact can improve behavior toward an ethnic minority and that network targeting can amplify effects.

7.2 Economic mechanism

The evidence points to affective engagement and empathy rather than stereotype correction. The documentary makes Santal lives salient and humanized, encouraging prosocial behavior.

7.3 Evidence chain

The evidence moves from information acquisition to prosocial games, emotional responses, spillovers, administrative disputes, and real-effort productivity.

7.4 Policy implications

Low-cost media interventions may complement direct contact and institutional policies, especially when combined with network targeting.

7.5 Caveats

The central-arm mechanism is not fully identified, long-run persistence is uncertain, and generalizability depends on audience selection and local interethnic context.

7.6 Final takeaway

Virtual contact can change intergroup behavior even when measured beliefs remain sticky; affective media and social networks together can foster practical social harmony.